Home Appliance Rental Marketplace

FINAL - Practical Non-Functional Requirements (NFR) and Core Business Rules | Version 3.0 | 13 October 2026

Scope: A simple, reliable, attractive rental marketplace - not an enterprise-level platform. Responsive web first; mobile-friendly design across phones, tablets, and laptops.

1. Project overview

One account can borrow appliances and list appliances for rent. Support multiple products, multiple lenders in a single checkout, daily/weekly/monthly rental plans, security deposits, third-party delivery, return inspections and dispute resolution.

2. Confirmed business rules

Rule	Final decision
Platform commission	Fixed 15% of product market value, per booked item/rental transaction.
Security deposit	Fixed 20% of product market value per item; refundable subject to inspection and valid deductions.
Multiple lenders	A single cart/checkout may include products from different lenders; create separate lender sub-orders.
Logistics	Delivery and pickup outsourced to a third-party delivery app/provider.
Duration	Daily, weekly and monthly rentals.
Damage and disputes	Lender inspects returned item and proposes evidenced deductions; admin reviews disputed claims.

3. Roles and essential screens

Role	Essential functions
Borrower	Browse/search/filter, product details, cart, dates, checkout, deposit tracking, active rentals, returns, dispute submission, profile.
Lender	List/edit items, upload photos, set daily/weekly/monthly rent, availability, booking approval, handover checklist, return inspection, earnings.
Admin	Dashboard, users, listing moderation, orders, refunds, dispute decisions, commission ledger, payout status, delivery exceptions.

4. Rental process

• Lender adds appliance with category, condition, market value, rental rates, photos, location/serviceability, deposit and availability.

• Borrower chooses items and rental dates; system checks availability for each item and reserves stock to avoid double booking.

• Cart shows itemized rent, 15% market-value commission treatment, 20% deposits, logistics fees and final payable amount before payment.

• Checkout creates one parent order and lender-specific sub-orders. Payment success must be confirmed by gateway webhook.

• Third-party delivery/pickup requests are tracked with reference IDs, status, failure handling and proof of handover.

• At return, lender uploads photos/checklist and proposes any damage or missing-item deduction. Borrower can accept or dispute.

• Undisputed eligible deposit is refunded; disputed amount is held for admin review while any undisputed balance is released under policy.

5. Money logic - critical decision to implement safely

Example: market value Rs 20,000, rent Rs 2,000: commission = Rs 3,000 (15% of market value), deposit = Rs 4,000 (20% of market value). The deposit is not platform revenue. Because commission exceeds rent, the checkout must explicitly show who pays the commission and obtain consent; the app must not silently deduct Rs 3,000 from a Rs 2,000 lender rental payout.

Before launch, approve whether the borrower pays commission as a separate fee, the lender pays it, or a defined split applies. Also finalize tax treatment, gateway charges, payout schedule, inspection deadline, refund timeline, cancellation fees and late-return policy. Do not promise instant refunds; actual settlement depends on payment provider.

6. Deposit and damage safeguards

• Record deposit separately from rental revenue; maintain a ledger of collection, hold, authorized deductions, refunds and payment references.

• Require timestamped pre-handover and post-return photos plus condition checklist; record accessories and serial number where applicable.

• Normal wear and tear is not automatically chargeable; publish a damage schedule and fair-evidence standard.

• Lender cannot unilaterally finalize disputed deductions. Admin reviews evidence, records reason and decision, and informs both parties.

• Use a compliant payment partner and approved fund flow for deposit collection/holding/refunds; do not assume the platform can operate its own escrow account.

7. Simple but polished UX/UI

• Modern, clean visual design with consistent spacing, strong contrast, readable typography, attractive appliance cards and clear product photos.

• Mobile-first responsive layouts for Android/iPhone, tablets, MacBook and desktop; touch targets at least 44px.

• Home: search bar, categories, location/service area, featured items; detail: photos, availability calendar, rental plans, deposit and total-cost breakdown.

• Checkout: simple steps, cart grouped by lender, transparent fee breakdown, clear confirmation and payment state.

• Borrower and lender dashboards share a consistent design but use role-specific navigation; visible rental status timeline and refund status.

• Accessibility: keyboard navigation, labels, error messages, sufficient contrast and understandable loading/empty states.

8. Practical non-functional targets

Area	Target / acceptance
Speed	Typical pages interactive within 3 seconds on a normal mobile connection; API p95 under 800 ms for ordinary requests at launch.
Availability	Target 99.5% monthly uptime excluding planned maintenance.
Security	HTTPS, hashed passwords, server-side role checks, validated uploads, secure payment webhooks, audit logs.
Reliability	No double booking; idempotent payment/refund processing; retry failed delivery updates safely.
Data	Daily encrypted backups; tested restore before launch; retain records under documented policy.
Privacy	Collect minimum personal data, role-restrict access, provide privacy policy and consent.
Compatibility	Latest two major Chrome, Safari, Firefox, Edge versions; responsive 360px to desktop widths.
Observability	Error logs, payment/refund alerts, admin issue dashboard and basic uptime monitoring.

9. Recommended affordable tech stack

Frontend: React + TypeScript + Tailwind CSS. Backend: FastAPI or Node.js (choose one). Database: PostgreSQL. Image storage: managed object storage. Payment: compliant Indian payment gateway. Delivery: third-party provider API if available, with manual admin fallback. Hosting: a small managed cloud/VPS initially; scale only when traffic requires it. Mobile apps can be added later if needed; a responsive web/PWA is sufficient for the first release.

10. Minimum database entities

User, Address, ApplianceListing, AppliancePhoto, AvailabilityBlock, RentalRate, Cart, Order, OrderItem, LenderSubOrder, Payment, DepositLedgerEntry, DeliveryJob, HandoverEvidence, ReturnInspection, DamageClaim, Dispute, Refund, Payout, Notification, AuditEvent.

11. MVP release checklist

• Borrower and lender registration/login and secure role-aware navigation work.

• Listing moderation, search, date selection and inventory conflict prevention work.

• Multi-lender checkout correctly splits orders and handles partial cancellations.

• 15% market-value commission and 20% market-value deposit calculate correctly per item, with explicit payer allocation.

• Payment webhook retries never create duplicate payments; refunds and payouts reconcile to ledger.

• Delivery integration or documented manual fallback, proof of handover and pickup work.

• Inspection evidence, borrower disputes, admin decisions and partial/full deposit refunds work.

• Mobile/tablet/desktop layouts, basic accessibility, backups and restore test pass.

12. Out of scope for first release

Advanced AI recommendations, dynamic pricing, complex insurance integrations, sophisticated fraud scoring, microservices, native iOS/Android apps and enterprise analytics are not necessary for launch.

Status: Core product scope and six user decisions confirmed. Fee payer, inspection time limits, tax/settlement mechanics and delivery provider contract remain launch-blocking policy decisions.